

XD Reporting

Insight Smart Calculation Builder

User Guide

Access Limited Distribution

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Document Control

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1. Overview

1.1. Purpose of the Guide

Calculations are used to define a new value / column to add to a report. These are a useful way to apply some initial data analysis on the report upfront.

Possible uses of a calculation

- Concatenate multiple fields into a single column.
- Project column data e.g. a date + 30 days, a value + 10% etc.

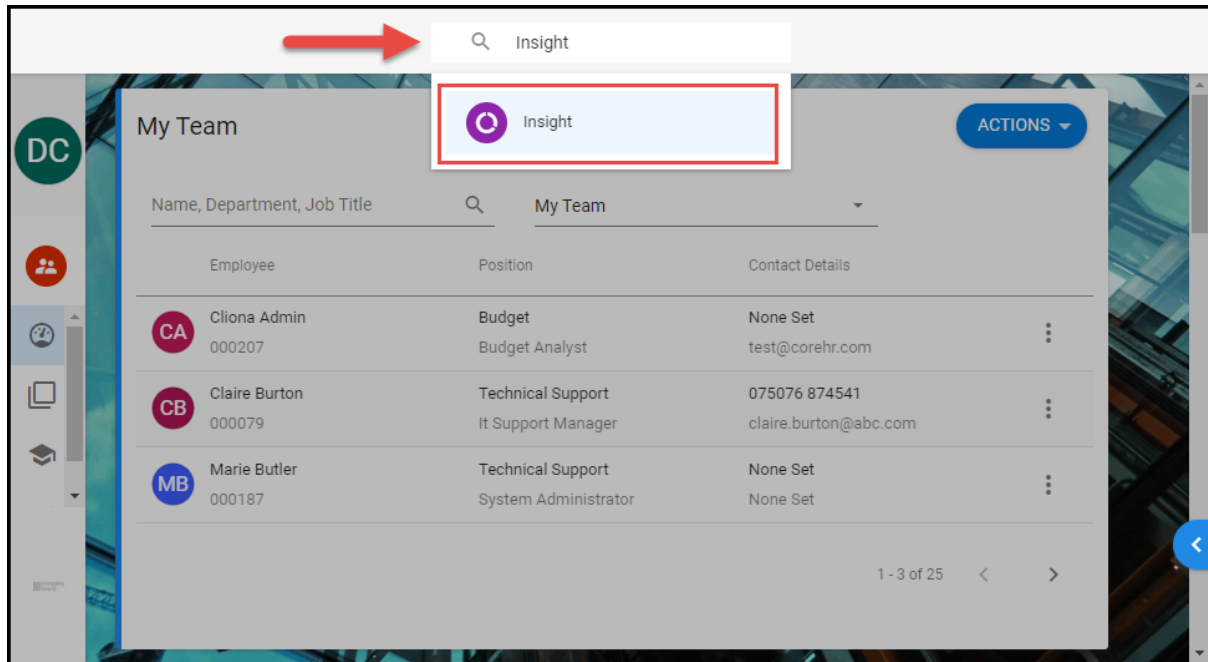
Once created, a calculation is effectively a new report column. As such it can be filtered on, used in a prompt, formatted, added to a chart etc.

2. Accessing the Smart Calculation Builder

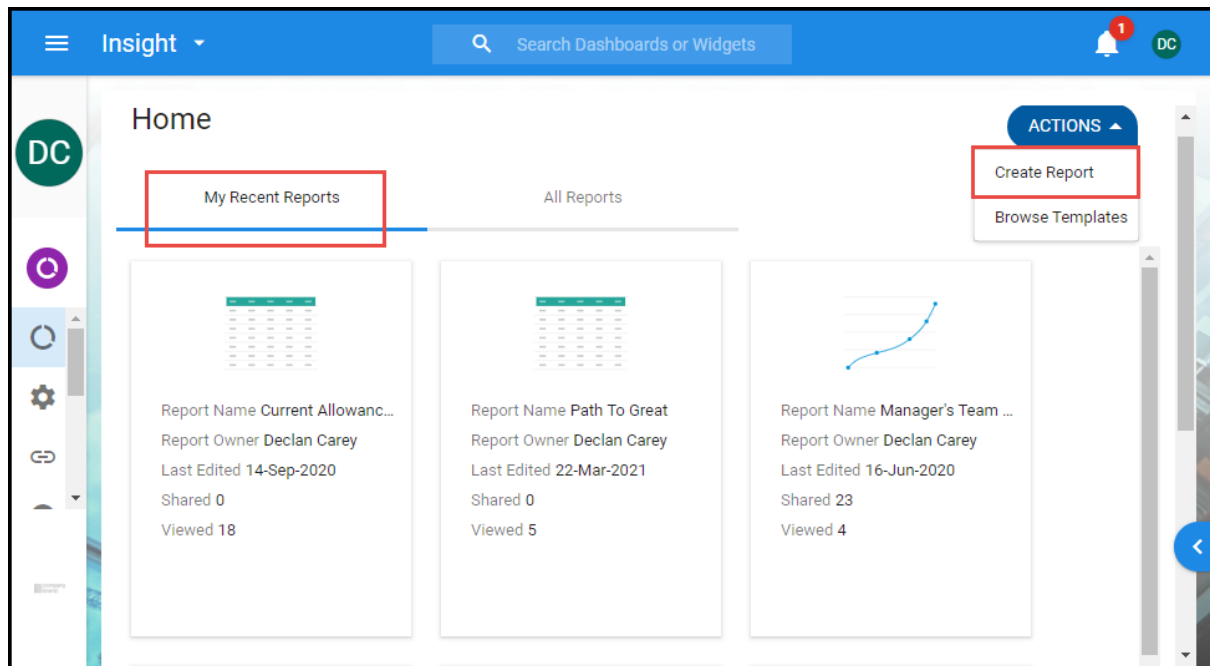
2.1. Creating a new Calculation

To access the Smart Calculation Builder for a new or existing Report on Insight, follow the below steps.

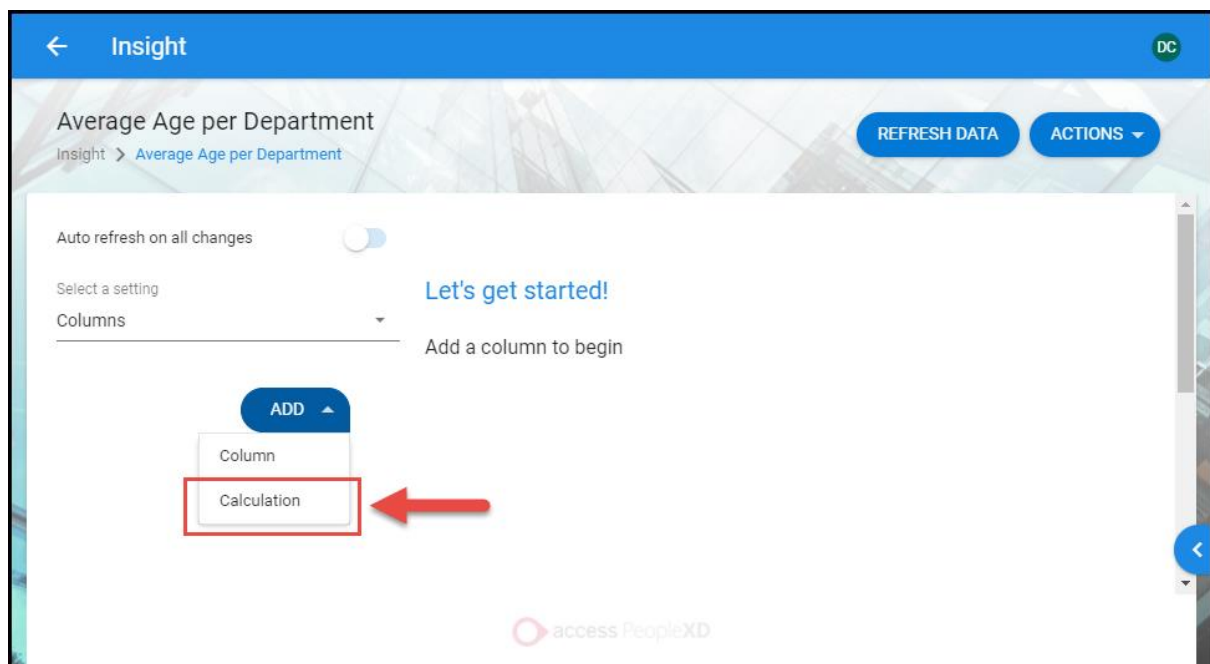
1. Quick Jump to Insight.



2. On the Home tab, select an existing report from the *My Recent Reports* tab or create a new report using the **Actions** button.



3. Within the *Edit Mode* of a Report, to open the new Calculation screen, click on the **Add** button and then the **Calculation** button.



4. The Smart Calculation Builder screen opens.

← Average Age per Department DC

Calculation

Insight > Average Age per Department* > Calculation ERROR CONSOLE

Favourite

Calculation Name* ☆

Data Fields Calculated Inputs XD Functions

Search 🔍

Hr Person ...	Department	⋮
Hr Person ...	Company	⋮
Hr Person ...	Age	⋮
Hr Person ...	Person Reference	⋮

Calculation Text

Type here or select data/functions from the left

access PeopleXD SAVE

5. Give the *Calculation* a Name.
 - a. This is used as the column header for the created calculation. Provide a meaningful description that makes it clear what this calculation is being used for.

2.2. Smart Calculation Builder Screen

The below table highlights the items that are shown on the *Smart Calculation Builder* screen.

Item	Description
Favourite	This allows a user to favourite a Calculation that can be quickly found again on the Calculated Inputs tab
Calculation Text	The Calculation text is automatically generated from the Calculated Inputs & XD Functions built into the software. The calculation can then be adjusted in this area.
Search	This allows a user to search for an items, fields, inputs, or functions on each tab.
Total Type	This gives the user the option to select from the following Total calculation items: • Average (numeric fields only) • Count (numeric fields only) • Sum (numeric fields only) • Max (numeric and date fields) • Min (numeric and date fields)
Total Label	This is the name of the Total type given on the report.

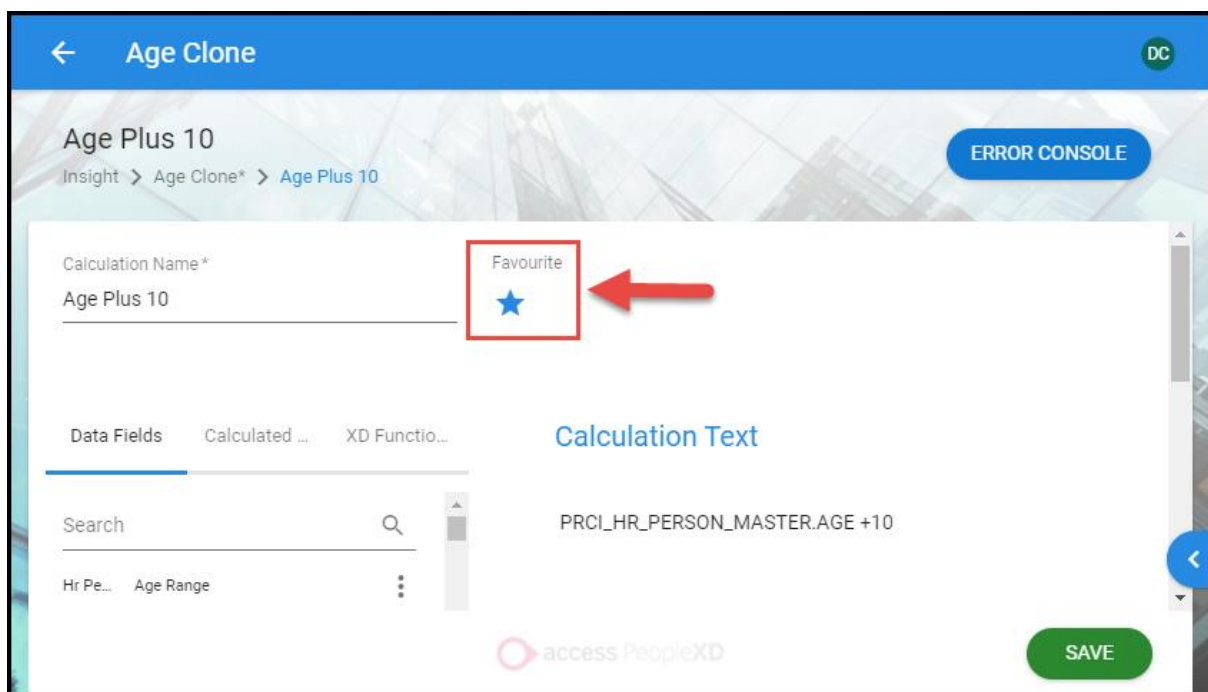
Item	Description
Hide Calculation	This hides the calculation column on the report once refreshed.
Error Console	This allows a user to access the Error Console easily within each calculation screen.

Note: The *Calculation Text* field is essentially your calculation builder. Depending on the type of calculation you require, you can construct a formula for this using a mix of the Data Fields, Calculated Inputs or XD Functions.

2.3. Adding a Calculation as a Favourite

To mark a calculation as a Favourite, follow the below steps

1. Open the Report in Edit Mode
2. Create a new Calculation or edit an existing calculation.
3. Click on the ★ button to mark this calculation as a favourite.



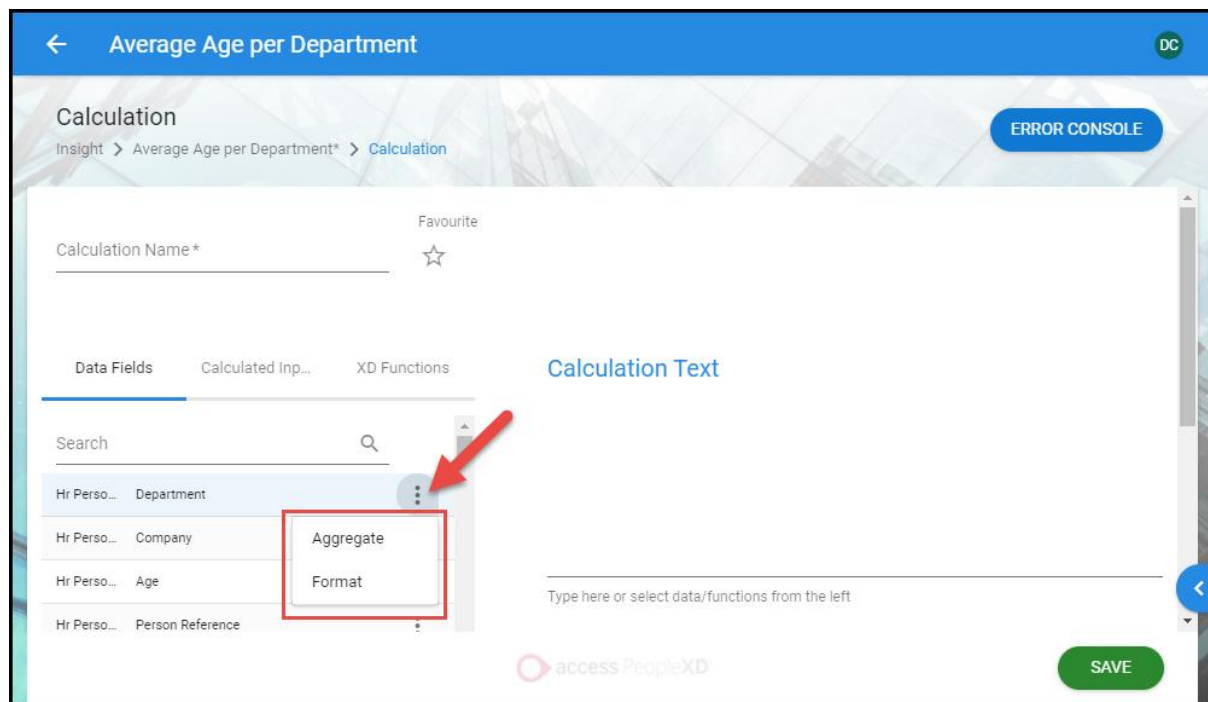
4. The software is smart enough to know when one is applicable to a report or not. Therefore, not all favourites are available to be selected on a report, avoiding possibilities of calculation falling over. The favourite refers to the views that can only be used on the report - it does not return a calculation that is not relevant to your dataset on the report.

3. Data Fields

3.1. Columns

The Data Fields section on the Smart Calculation Builder behaves in the same way as *Columns* do on tab reports in the *Report Wizard*. However, there are two additional settings available for these data fields.

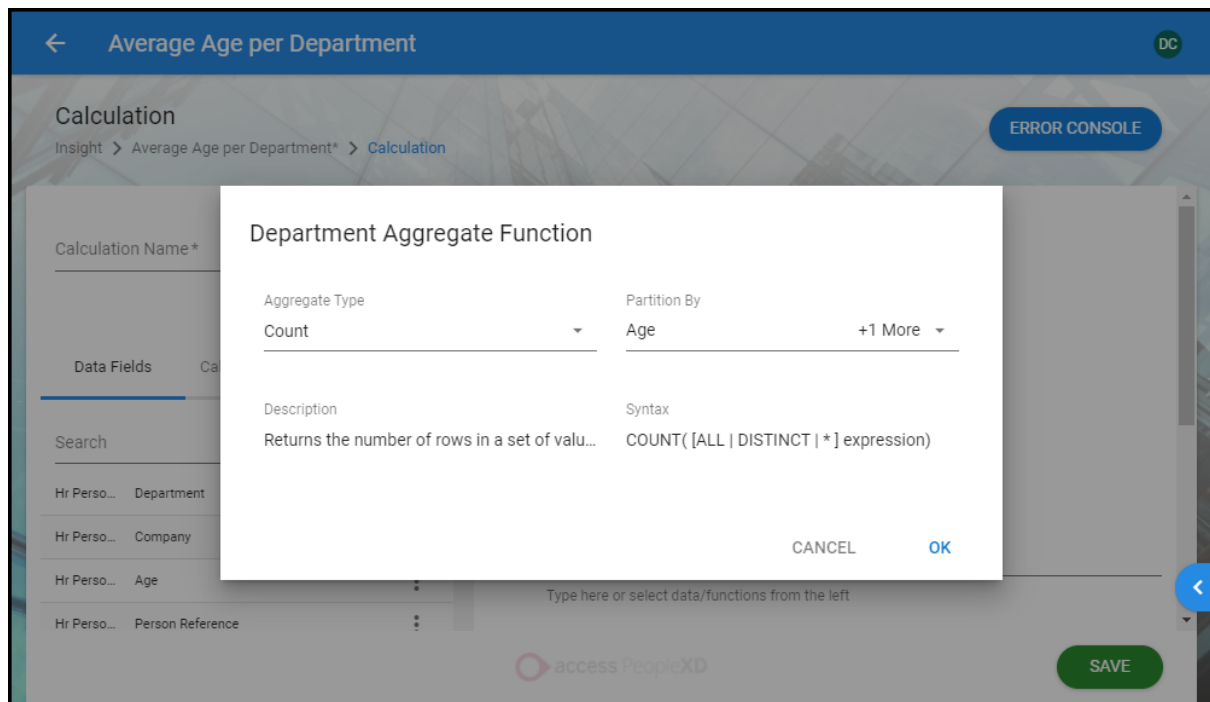
1. Aggregate
2. Format



3.1.1. Enable Aggregate Functions

Aggregate functions can be added to report columns at the point of adding the column to the report from the *Smart Calculation Builder*.

The following image and table outlines the details on the Aggregate Function screen.

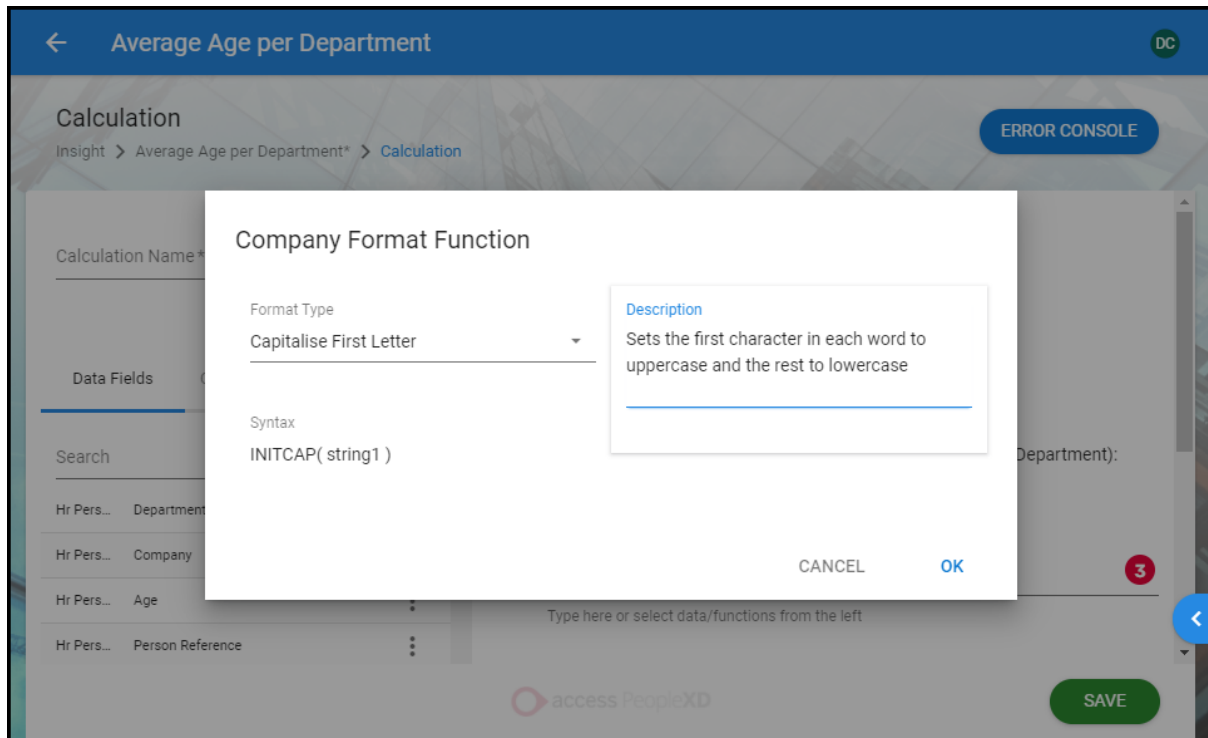


Item	Description
Aggregate Type	The aggregate options available are: • Average (numeric fields only) • Count (numeric fields only) • Sum (numeric fields only) • Max (numeric and date fields) • Min (numeric and date fields)
Partition By	The Partition By clause divides the result set into partitions and changes how the window function is calculated. It does not reduce the number of rows returned.
Description	This gives a brief explanation of what this aggregate function is doing

Item	Description
Syntax	This shows the calculation text / code that is being called. This is shown on the <i>Calculated Text</i> area once added to the Calculation.

3.1.2. Format

This allows you to add some formatting rules to the calculation when it appears on the Report. The following image and table outlines the details available on this screen.



Item	Description
Format Type	The following options are available to format a calculation by: • Capitalise First Letter • Convert to Character • Convert to Date • Left Pad • Lowercase • Replace Character • Right Pad • Round Number

Item	Description
	• Substring • Upper Case
Padding Length	Padding data or in other words adding a specific set of characters to the left of the string or right of the string can be done using inbuilt T-SQL and PL/SQL functions.
Padding Character	Padding characters are characters used to fill up remaining available character spaces in a field. For example, if a field requires a minimum of 15 characters.
Description	This gives a brief explanation of what this format option is doing
Syntax	This shows the calculation text / code that is being called. This is shown on the <i>Calculated Text</i> area once added to the Calculation.

4. Calculated Inputs

4.1. Input Types

The Calculated Inputs consist of the following types of items that can included on a Report.

1. Calculations (shown with Green icon)
2. Favourites (shown with Yellow icon)
3. Prompts (shown with Teal icon)

The following image shows the breakdown of these

The screenshot shows the 'Average Age per Department' calculation builder interface. The top header is blue with a back arrow, the title 'Average Age per Department', and a 'DC' indicator. Below the header, the main area is titled 'Calculation' and includes a breadcrumb 'Insight > Average Age per Department* > Calculation' and an 'ERROR CONSOLE' button. The interface is divided into two main sections. On the left, there is a 'Calculation Name*' input field and a 'Favourite' star icon. Below this is a tabbed interface with 'Data Fields', 'Calculate...', and 'XD Functi...'. The 'Calculate...' tab is active, showing a search bar and a list of input types: 'Average Age by Depa...' (green icon), 'Enter Department Na...' (yellow icon), and 'Age Plus 10' (teal icon) with a 'VIEW' button. On the right, there is a 'Calculation Text' input area with a placeholder 'Type here or select data/functions from the left' and a character count '0/300'. At the bottom right, there is a 'SAVE' button. The 'access PeopleXD' logo is visible at the bottom center.

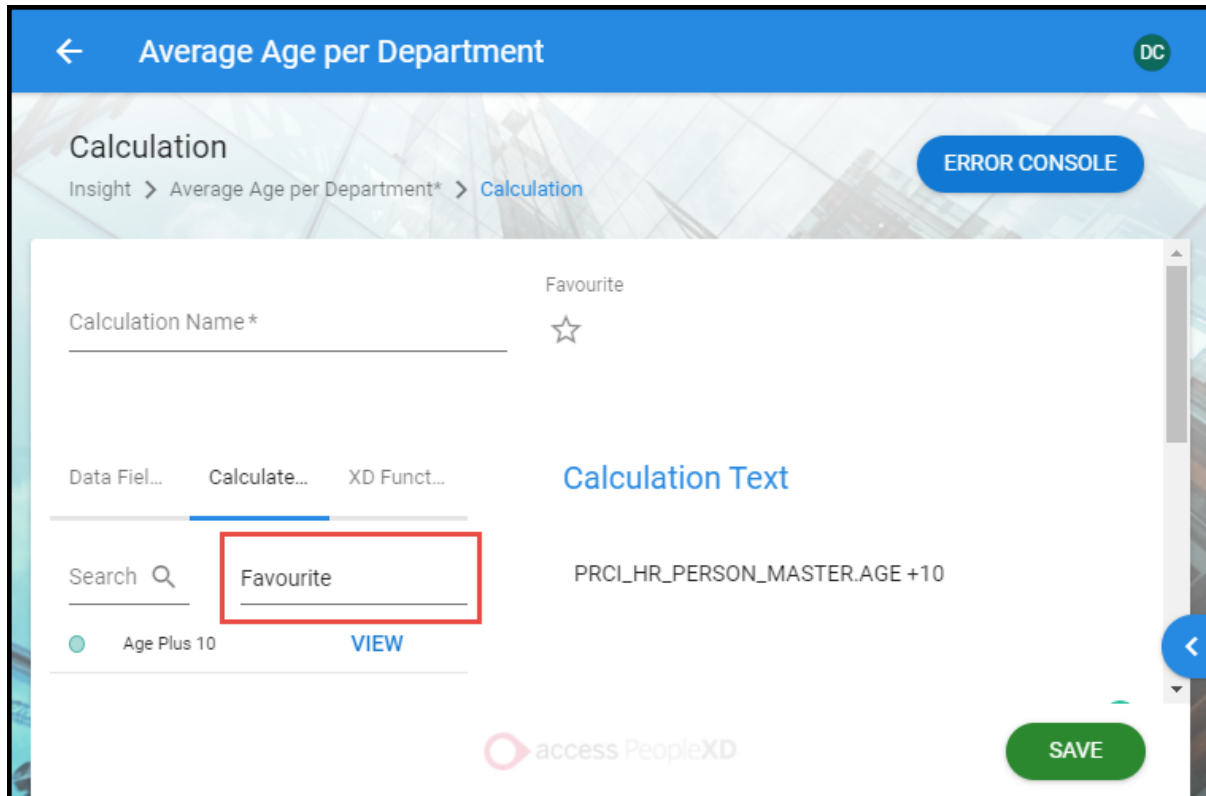
4.2. Calculations

Calculations that have been created specifically for this Report show up on this screen. The image below shows the calculation appearing once filtered correctly.

The screenshot shows the 'Average Age per Department' calculation builder interface. At the top, there is a blue header bar with a back arrow, the title 'Average Age per Department', and a 'DC' indicator. Below the header, the main content area is titled 'Calculation' and includes a breadcrumb trail: 'Insight > Average Age per Department* > Calculation'. A blue 'ERROR CONSOLE' button is located in the top right corner. The interface is divided into two main sections. On the left, there is a 'Calculation Name*' input field with a 'Favourite' star icon to its right. Below this, there are three tabs: 'Data Fields', 'Calculate...', and 'XD Functi...'. The 'Calculate...' tab is active. Under the 'Calculate...' tab, there is a search bar with a magnifying glass icon. Below the search bar, there is a list of calculations. One calculation, 'Average Age by Depa...', is highlighted with a red box. To the right of the search bar, there is a 'Calculation Text' section. It contains the text ':CALC(Average Age by Department): :CALC(Average Age by Department):'. At the bottom of the interface, there is a 'SAVE' button and a 'access PeopleXD' logo.

4.3. Favourites

The image below shows the favourited calculation appearing once filtered correctly. The software is smart enough to know when one is applicable to a report or not. Therefore, not all favourites are available to be selected on a report, avoiding possibilities of calculation falling over.



4.4. Filter Prompts

Filters are used to help restrict or limit the data returned in order to create a more meaningful report rather than a straightforward data extract for example

- All employees in *Department X*.
- All employees with a *Date Joined* in 2019.

Many filters can be applied to a report.

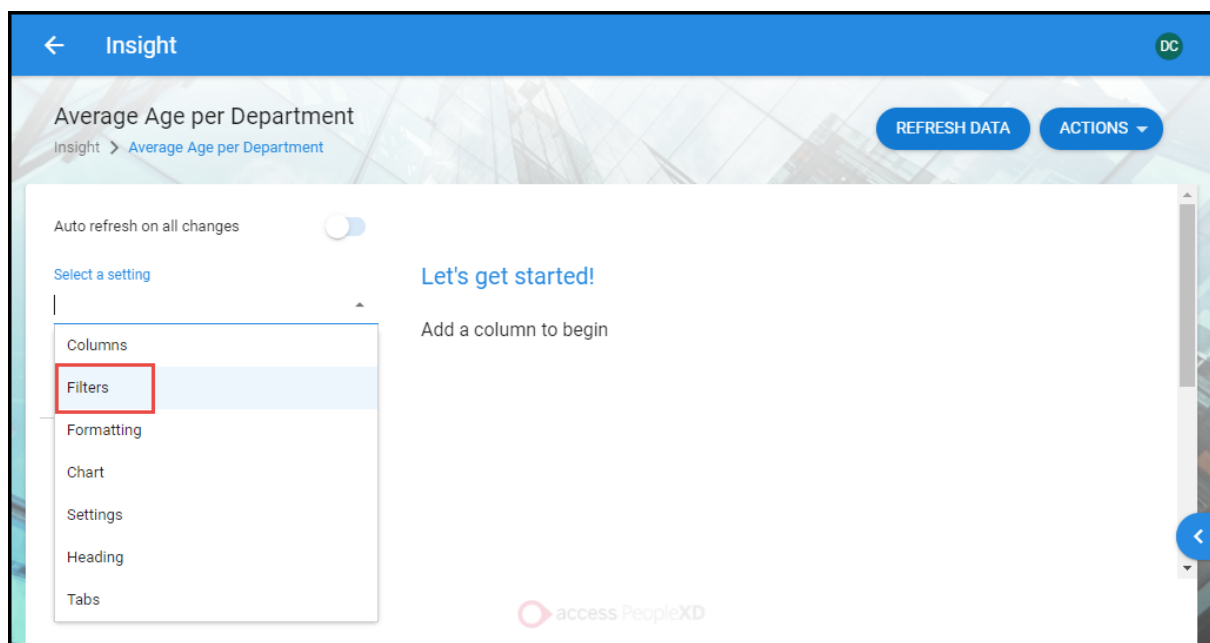
You can create nested filters. For example, a filter could be used to return all employees that left in January 2019, using a filter on their leave date:

- On or after 1st Jan 2019
- On or before 31st Jan 2019

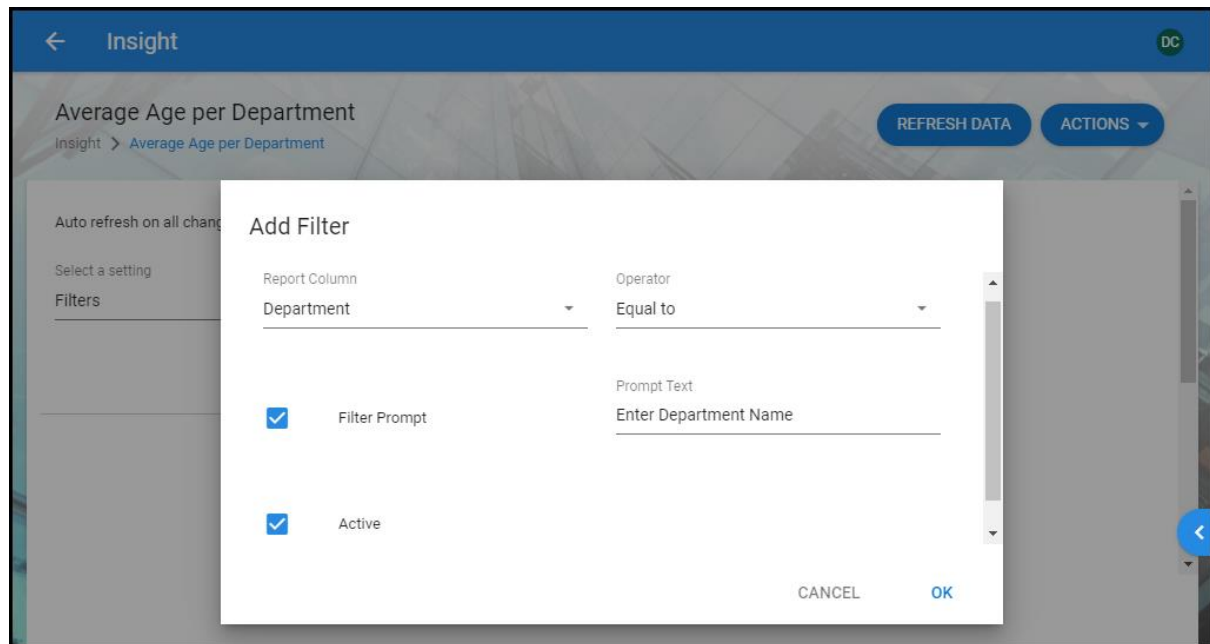
4.4.1. Creating a Filter

Follow these steps to add a filter to a report:

1. Enter *Edit Mode* for a Report
2. Select **Filters Setting** and click **Add**.

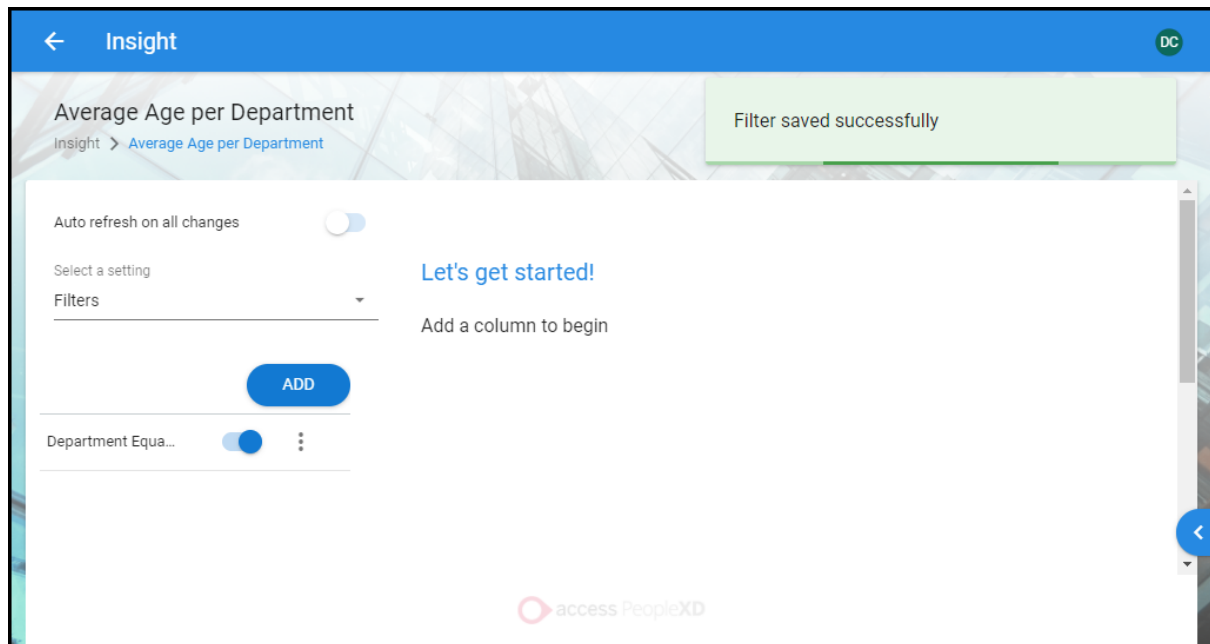


3. The *Add Filter* screen opens.



4. Select the Report column from the report to filter on.
 - a. This list displays columns that are included on the report at the top. If you need to filter based on a column not included in the report, other columns are available to be selected on the list. That way you can apply a filter to it, but it does not appear on the report.
5. Select the Operator
 - a. This is how you filter on the select column e.g. where something is equal to, less than, after etc. Depending on the column selected the Operator list differs. This is because there are different operators for different data types e.g. Text, Numeric or Date.
 - b. If using the options "In the list" or "Not in the list", you need to provide a comma separated list of items in Filter Text to compare against.
3. Select a *Filter Prompt* if required. See Filter Prompts for further details.
4. Ensure **Active** is selected.
5. Click **OK** to save.

The filter is created and listed on the *Filter* screen.

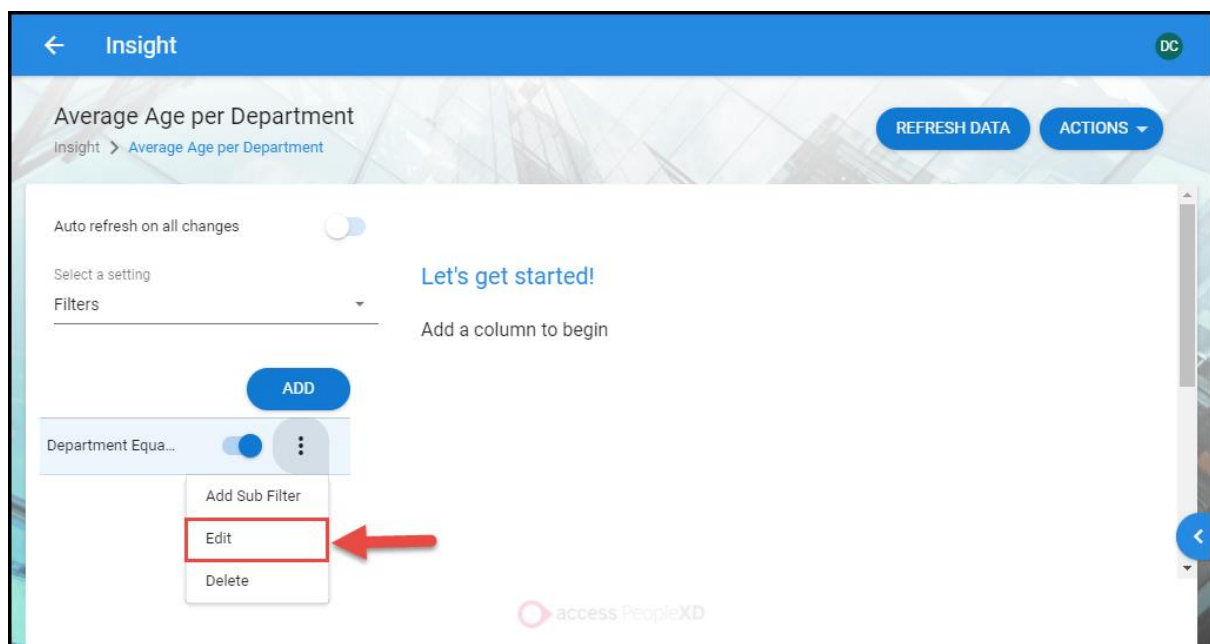


4.4.2. Creating a Prompt

Prompts can be added to a filter. This means that when the report is run, a prompt is shown on screen asking for the value to filter by e.g. "Enter Year" or "Select the Pay Period". This approach is useful as the report is dynamic and can be run to display different data sets every time, rather than pre-filtering of specific values.

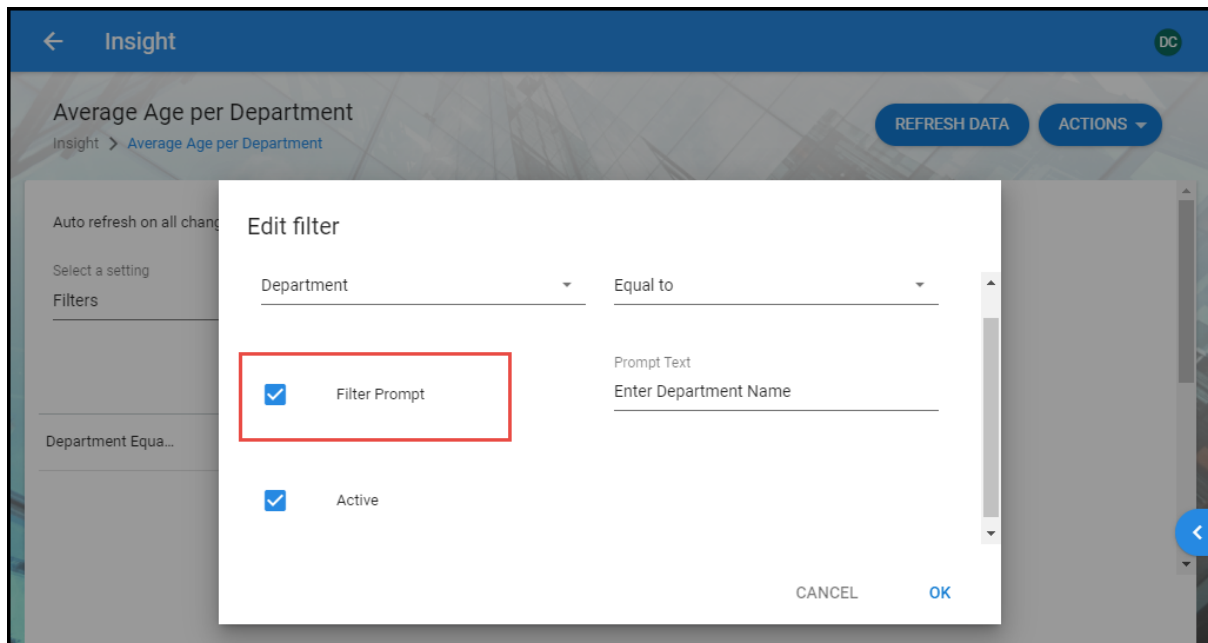
Follow these steps to add a prompt to your report.

1. On the Filter screen, click on the Edit button against the selected Filter.



2. Select **Filter Prompt**.

- a. Enable this to ensure that a prompt appears when the report is run, requesting you to input the required value.



3. Provide the *Prompt Text*.

- a. This is the message that appears to the user when running the report guiding them to provide the value.

4. Ensure **Active** is selected.

5. Click **OK**. The filter is saved automatically.

6. Filters that include prompts can be identified on the *Filters* screen as having the *Prompt Text* in brackets.

Average Age per Department

Insight > Average Age per Department

Auto refresh on all changes ☐

Select a setting

Filters

Department Equal to (Enter Department Name)

ADD

Note:

- Prompts are not mandatory. Therefore, you can ignore the report and run for all data by clicking **OK**.
- Filters that include prompts can be identified on the *Filters* screen as having the *Prompt Text* in brackets
- You can setup multiple prompts for a report. If more than one active filter includes a prompt, this appears as follows when you run the report:

4.4.3. Select Prompt

The image below shows the prompt appearing once filtered correctly.

← Average Age per Department DC

Calculation

Insight > Average Age per Department* > Calculation ERROR CONSOLE

Calculation Name* Favourite

Data Fields Calculate... XD Functi...

Search

Enter Department Na...

Calculation Text

:PROMPT(Enter Department Name):

access PeopleXD SAVE

5. XD Functions

5.1. What are XD Functions?

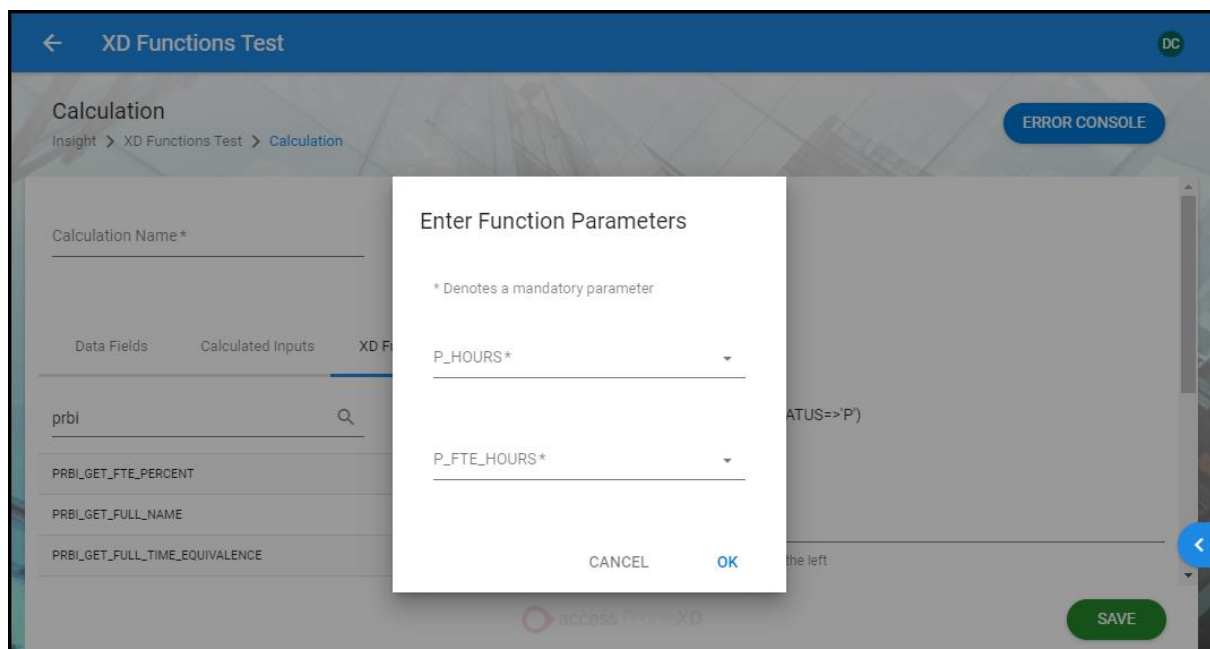
These are standard functions built into Insight to perform operations on multiple values to create summary results based on XD data fields. When selecting XD functions to use on a report, it enables you to define logical columns together with complex expressions and syntax code.

There is a wide variety of existing functions within XD that are supported here and are built from XD Business Areas and Data Fields within Insight.

5.2. Enter Function Parameters

On each XD function created, the parameters returned are relevant and specific to the function called. For example:

- P_empno
- P_date
- P_fte_hours



5.2.1. Order of items

Items from the function are added and then the calculation text box can be adjusted with further additional functions or manually entered ones.

The screenshot displays the 'XD Functions Test' interface. At the top, there's a blue header with a back arrow and the title 'XD Functions Test'. Below the header, the 'Calculation' section is active, showing a breadcrumb trail: 'Insight > XD Functions Test > Calculation'. An 'ERROR CONSOLE' button is visible in the top right. The main area is divided into three tabs: 'Data Fields', 'Calculated Inputs', and 'XD Functions'. The 'XD Functions' tab is selected, showing a search bar with 'prbi' and a list of functions: 'PRBL_GET_FTE_PERCENT', 'PRBL_GET_FULL_NAME', and 'PRBL_GET_FULL_TIME_EQUIVALENCE'. A 'Favourite' star icon is next to the search bar. On the right, the 'Calculation Text' box is highlighted with a red border, containing the formula: 'PRBL_HR_FTE_IND(P_PERSON_STATUS=>'P') PRBL_GET_FTE_PERCENT(P_HOURS=>'7', P_FTE_HOURS=>'37.5')'. Below the formula, there's a placeholder text 'Type here or select data/functions from the left' and a character count '93/300'. A 'SAVE' button is at the bottom right. The 'access PeopleXD' logo is at the bottom center.

6. Error Handling

6.1. Error message shown on calculation text

If a report generates a 404 error on screen when it is generating, the *Error Console* can be used to help identify the cause.

6.2. Error console

The error console for the report provides the following information that can be helpful when troubleshooting the cause of the error:

- Who ran the report?
- The date and time it was run.
- The error message.
- The SQL used to build the report.

If there are multiple errors generated, these are listed on screen.